Satellite Topographic Mapping in Complex Intertidal Wetlands: A Canopy-Height-Constrained Fusion of ICESat-2 Photon-Derived Structural Samples and Single-Phase Submeter Optical Imagery

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***Abstract*—High-resolution topographic mapping of intertidal wetlands is essential for geomorphic analysis, yet existing remote sensing methods often struggle with vegetation interference, dependence on dense time-series data, and limited representation of fine geomorphic features. We propose a canopy-height-constrained stratified cooperative inversion framework for the entire intertidal wetland, integrating single-phase submeter optical imagery (Jilin-1), spaceborne photon-counting light detection and ranging (LiDAR) Ice, Cloud, and Land Elevation Satellite 2 (ICESat-2), and machine learning. To accurately construct digital elevation model (DEM) and canopy height model (CHM) training samples in salt-marsh environments, we developed an ATL03 photon-classification workflow combining histogram-based control-point extraction and morphological refinement to generate these samples directly from ICESat-2 ATL03 photons. The retrieved CHM was then introduced as a structural constraint in the DEM retrieval model to support canopy-terrain signal decoupling in vegetated salt-marsh areas. A case study on Chongming Island, Shanghai, China, demonstrated that the DEM retrieval achieved high accuracy on the test set (R^2 = 0.94, root-mean-squared error (RMSE) = 0.28 m) and maintained consistent performance against independent UAV-LiDAR validation data (R^2 = 0.53 – 0.77 and RMSE = 0.34–0.53 m). The retrieved 0.5-m DEM reproduced regional elevation gradients, tidal-creek networks, and microtopographic variations across bare flats and vegetated marshes. Shapley additive explanation (SHAP) analysis showed that elevation retrieval over bare mudflats relied mainly on spectral predictors, whereas vegetated areas exhibited a complementary spectral-texture-CHM structure, with CHM consistently ranking as a mid-to-high predictor (fourth–seventh). This further supports the role of CHM as an effective structural constraint. By using only single-phase imagery and ATL03-derived DEM/CHM samples, the framework enables intertidal topographic retrieval that includes vegetated areas. It therefore provides an efficient and low-cost pathway for high-accuracy intertidal topographic monitoring under complex environmental conditions and limited image availability.**

***Index Terms*—Elevation retrieval, Ice, Cloud, and Land Elevation Satellite 2 (ICESat-2), Jilin-1, machine learning, multisource remote sensing, UAV-light detection and ranging (LiDAR).**

I. Introduction

THE intertidal zone is an extensive, low sedimentary landform between the mean low water and the mean high water levels of spring tides. Broadly, it includes bare mudflats, tidal creeks, salt marshes, and mangroves [1]. As a land–sea ecotone, it plays a significant role in wave dissipation and coastal protection [2], blue carbon sequestration [3], and biodiversity conservation [4]. However, driven by human activities and sea-level rise, global intertidal ecosystems have declined markedly in recent decades: salt marshes lost 1452.84 km^2 (2000–2019), mangroves decreased by 3363 km^2 (2000–2016), and bare mudflats potentially shrank by >20 000 km^2 (1984–2016) [5], [6], [7]. Such degradation reduces ecosystem stability and elevates coastal hazard risk [8]. Intertidal topography, governed by sediment supply, hydrodynamics, and biogeochemical processes, is a key indicator of future erosion risk and resilience [9]. Therefore, large-scale, high-precision intertidal digital elevation models (DEMs) are needed to quantify landform evolution and to support coastal research and management.

Currently, intertidal elevation measurement techniques can be classified into four primary categories: contact measurement, near-surface remote sensing, aerial remote sensing, and satellite remote sensing. Contact methods (e.g., real-time

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kinematic) provide point-wise in situ elevations [10]. Near-surface sonar or laser systems on vessels or tripods map topography over limited ranges [11]. However, both require manual operation or nearshore deployment and are often hindered by tidal inundation, dense vegetation, muddy terrain, and weather variability. Aircraft-borne passive or active sensors, operating at hundreds of meters altitude, can produce local DEMs with high accuracy and flexible operation, but coverage and cost remain limiting factors [12].

Satellite remote sensing, offering global coverage and regular revisits, underpins large-scale intertidal elevation mapping [13]. Existing satellite-based DEM retrieval methods have improved regional-scale intertidal topographic mapping through different data sources and retrieval strategies, yet their applicability remains constrained in complex wetland environments. Stereopair-based methods estimate elevation from image parallax, but their applicability is limited in low-relief, weakly textured, and tidally affected intertidal environments [14]. Interferometric synthetic aperture radar (InSAR) retrieves surface elevation from radar phase information, but wet surfaces and dynamic intertidal environments may cause signal attenuation and coherence loss [15], [16]. Waterline and inundation-frequency methods are widely used for intertidal DEM retrieval because they infer elevation from repeated observations of water-land boundaries or inundation states under different tidal stages [17], [18], [19]. However, these two types of methods require large volumes of multitemporal observations, and their performance is constrained by the number of valid observations, tidal coverage, and elevation-control data in practical applications. Their reliability may also decrease when waterlines or inundation boundaries are obscured by vegetation canopies. Spaceborne light detection and ranging (LiDAR), represented by Ice, Cloud, and Land Elevation Satellite 2 (ICESat-2), provides decimeter-level photon elevations and partial penetration capability through water and vegetation [20]. Nevertheless, sparse along-track sampling makes it difficult to reconstruct continuous DEM surfaces from ICESat-2 alone.

Building on the high-accuracy but spatially sparse nature of ICESat-2 photon elevations, recent studies have combined them with image-derived predictors and machine learning models to generate continuous DEM surfaces [21], [22]. Various algorithms, including RF [23], XGBoost [24], LightGBM [25], and SVM [26], have been used for this purpose. However, these methods remain limited in vegetated areas. Existing image-derived features mainly provide an indirect representation of ground elevation through canopy-surface characteristics and have limited ability to decouple canopy signals from terrain signals. This limitation arises because image-derived signals are easily affected by canopy obstruction and canopy-ground mixing. ICESat-2 provides photon-level vertical structure information and therefore offers the potential to separate vegetation and terrain signals at the photon scale. However, the low stature and dense structure of salt-marsh vegetation make it difficult to distinguish ground photons from vegetation photons in ATL03 data, although this distinction is essential for constructing reliable DEM and canopy height model (CHM) samples. In addition, most existing satellite-based intertidal DEM retrieval methods rely on medium-resolution imagery, typically at 10–30-m resolution, with limited attempts at high-resolution retrieval. This limits their ability to resolve tidal-creek networks and micro-topographic variations. These gaps highlight the need for an intertidal DEM retrieval framework that can better account for canopy-terrain signal decoupling in vegetated areas, reduce dependence on dense long-term image sequences, and support high-resolution topographic retrieval.

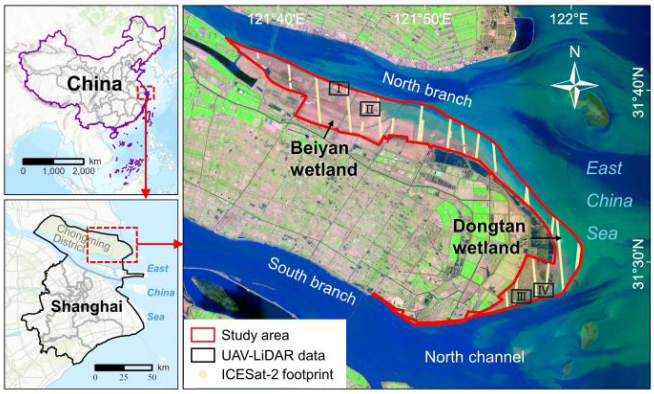


Fig. 1. Overview of the study site and data sources.

In this study, we developed a canopy-height-constrained intertidal DEM retrieval framework that integrates single-phase 0.5-m optical imagery (Jilin-1), ICESat-2 ATL03 photon data, and machine learning. The framework is built on two linked components: ATL03-derived DEM/CHM sample construction and CHM-constrained DEM retrieval. First, we developed a photon denoising and classification workflow that combines histogram-based control-point extraction, statistical filtering, and morphological refinement to identify and extract ground and vegetation photons in complex intertidal environments. This process generates high-quality DEM and CHM training samples directly from ICESat-2 ATL03 photons. Second, the retrieved CHM is introduced as a structural constraint in the DEM retrieval model to support canopy-terrain signal decoupling in vegetated salt-marsh areas. By combining ATL03-derived structural samples with single-phase high-resolution imagery, the proposed framework aims to enable fine-scale intertidal topographic retrieval across bare flats and vegetated marshes while reducing dependence on dense long-term optical image sequences.

II. Materials and Data

A. Study Area

Chongming Island, located at the confluence of the Yangtze River Estuary in Shanghai, China, is the world's largest alluvial island. The study area focuses on the northern and eastern sectors of the island (Fig. 1), specifically encompassing the Beiyan Wetland and Dongtan Wetland (31.44°N–31.71°N and 121.61°E–122.02°E). As a critical ecological buffer zone, this region sustains exceptional biodiversity and serves as a vital migratory corridor and wintering ground for avian species. Notably, the Dongtan Wetland holds international recognition as a Ramsar Site of Ecological Significance [27]. However, the estuarine system faces unprecedented challenges due to a marked reduction in sediment influx from the Yangtze River

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TABLE I
INFORMATION OF MULTISOURCE DATA

Data platform	Acquisition time	Parameters
JL1-KF02	2024-07-29	Spatial resolution: 0.5–2 m Band1: Blue (450–510 nm) Band2: Green (510–580 nm) Band3: Red (630–690 nm) Band4: NIR (770–895 nm)
ICESat-2 (ATL03, V7)	2024-01-28 2024-02-11 2024-03-11 2024-03-30 2024-04-28 2024-06-29 2024-07-28 2024-08-11 2024-08-30 2024-09-09 2024-09-28 2024-12-28	Flight altitude: 500 km Pulse spacing (along-track): ~0.7 m Beam spacing (intra-pair): ~90 m Beam spacing (inter-pair): ~3.3 km Laser wavelength: Green
UAV-LiDAR (DJI M300 RTK + Zenmuse L2)	2024-12	Flight velocity: 15 m/s Flight altitude: 100 m Ranging accuracy: < 5 cm Side overlap: 20 % Laser wavelength: NIR
UAV-LiDAR (V10 + RIEGL VUX-1LR[22])	2025-04	Maximum range: 1630 m Ranging accuracy: 1.5 cm Side overlap: 35% Laser wavelength: NIR

and accelerating global sea-level rise, collectively driving a transition from net depositional to erosional regimes [28], [29], [30]. This environmental shift poses substantial risks to regional ecological stability. Consequently, acquiring high-precision DEM data has become a priority for this vulnerable ecosystem, enabling detailed analysis of hydrodynamic processes and facilitating informed conservation strategies to address the estuary's ecological evolution.

B. Jilin-1 MS Imagery

High-spatial-resolution imagery for the study area was obtained from the Jilin-1 KuanFu 02 (JL1-KF02) satellite on July 29, 2024 (Table I). As a submeter optical satellite, JL1-KF02 provides a 150-km swath width and supports multiple imaging modes, including push-broom, inertial space, and multitarget imaging. The onboard sensor consists of one panchromatic (PAN) band and four multispectral (MS) bands: blue (450–510 nm), green (510–580 nm), red (630–690 nm), and near infrared (NIR, 770–895 nm). The PAN band offers a spatial resolution of 0.5 m, while the MS bands have a native resolution of 2 m. Further technical specifications can be found on the satellite's official website (https://www.jl1.cn/). To prepare the imagery for analysis, we performed geometric correction, radiometric calibration, and atmospheric correction to ensure geometric accuracy and derive surface reflectance suitable for quantitative applications. Subsequently, pan-sharpening was applied using the PAN band, resulting in a four-band MS dataset at a uniform spatial resolution of 0.5 m for all subsequent analyses.

C. Satellite Altimetry Data

In this study, we utilized the Level-2A Global Geolocated Photon Data (ATL03, Version 7), which provides acquisition time, latitude, longitude, and elevation for each detected photon. These data are publicly available from the National Snow and Ice Data Center (NSIDC). We collected all ATL03 acquisitions transecting the study area in 2024, comprising 12 ground tracks (Fig. 1 and Table I). Photon information was extracted using PhoREAL v3.30. Given their higher signal-to-noise ratios and superior vertical precision, we retained only strong beams for analysis. For DEM sample construction, classified ground photons from all available 2024 ATL03 acquisitions were used. For CHM sample construction, classified vegetation photons were restricted to June–September 2024 to better match the satellite-image acquisition period and the peak growth stage of salt-marsh vegetation. For each vegetation photon, we identified the nearest along-track ground photon within the same beam and calculated canopy height as the elevation difference between the paired vegetation and ground photons. All photons were spatially coregistered to the optical imagery grid. For photons falling within the same pixel, we aggregated CHM values using the 95th percentile height and averaged ground elevations for the DEM. After quality control, we retained 5883 CHM samples and 65 008 DEM samples.

D. UAV-LiDAR Validation and CHM Measured Data

To independently validate the accuracy of the retrieved DEM, we utilized UAV-LiDAR data collected from four representative sites (Fig. 1 and Table I). For Sites I and II, data were acquired in December 2024 using a DJI Matrice 300 RTK platform equipped with a DJI Zenmuse L2 LiDAR sensor, with a 60% side overlap. For Sites III and IV, data were collected in April 2025 using a V10 UAV platform equipped with a RIEGL VUX-1LR22 sensor, with a 35% side overlap. Ground points were extracted using a three-stage progressive filtering strategy: 1) removal of gross outliers based on predefined elevation constraints; 2) initial ground separation using the cloth simulation filtering (CSF) algorithm [31]; and 3) manual inspection to remove residual noise from the classified ground points. Based on the resulting high-confidence ground points, we generated DEMs via triangulated irregular network interpolation in ArcGIS Pro and subsequently converted the DEM surfaces to vector points. This process yielded validation samples of 158 370, 150 824, 193 301, and 294 087 for Sites I–IV, respectively. In addition, field-measured vegetation heights collected from June to September 2024 were used as reference data to validate the CHM inversion results.

III. METHODS

As shown in Fig. 2, the workflow consists of three main stages. First, we preprocessed the Jilin-1 imagery and engineered a multidimensional feature dataset. In parallel, we performed photon denoising and classification on ATL03 data to extract canopy height and ground elevation information. Second, we implemented a stratified cooperative retrieval strategy. We first trained a canopy-height estimation model using ICESat-2 canopy heights as the target variable together with

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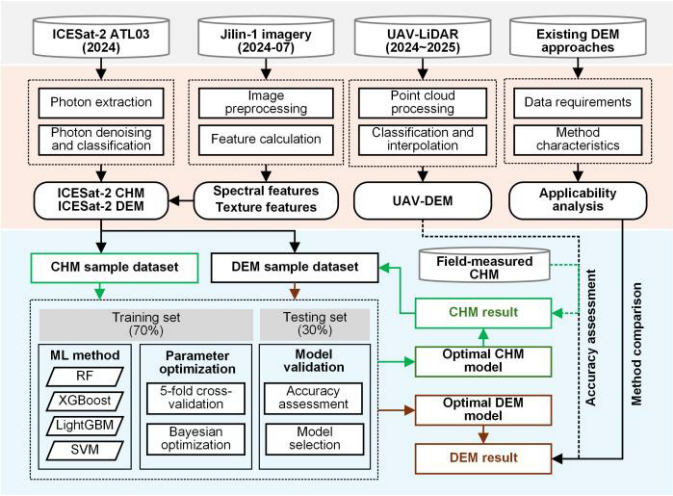


Fig. 2. Overall workflow of the proposed method.

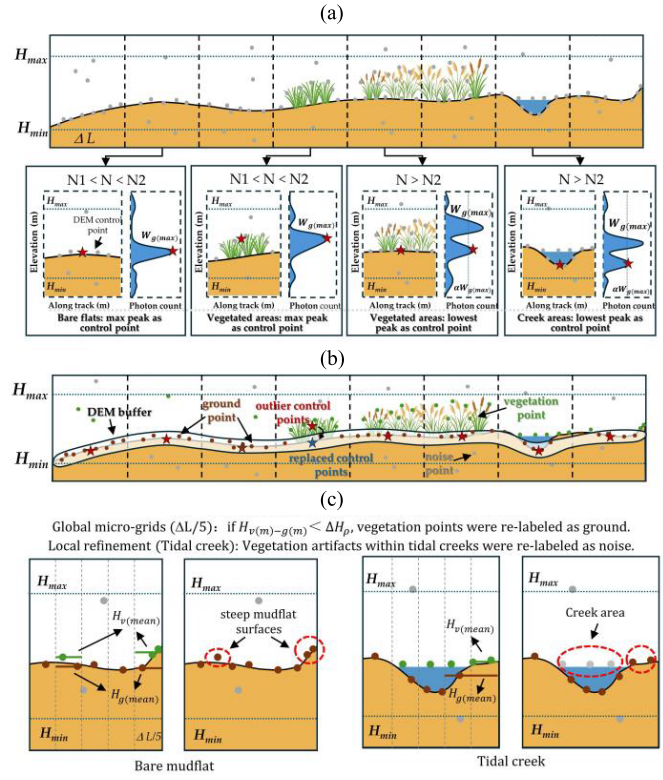


Fig. 3. Workflow of ICESat-2 photon denoising and classification. (a) DEM control-point extraction based on multipeak histogram. (b) DEM control-point correction and photon classification. (c) Refinement of classification results.

the imagery feature set. We then developed a DEM retrieval model based on ICESat-2 ground photons by jointly using the imagery features and the previously generated CHM as a key structural predictor. Third, we applied the selected model to the entire study area to generate wall-to-wall elevation maps and assessed mapping accuracy using UAV-LiDAR elevation data.

A. ICESat-2 Photon Denoising and Classification

To classify ATL03 photons over complex intertidal environments, we proposed a robust workflow that combines histogram-based control-point extraction, statistical filtering, and morphological refinement. The method partitions photons into three classes (ground, vegetation, and noise) in three steps (Fig. 3).

1) DEM Control-Point Extraction Based on Multipeak Histogram: Unreliable photons (Confidence ≤ 0) were filtered using ICESat-2 confidence flags, and a global elevation constraint $[H_{min}, H_{max}]$ derived from prior terrain data was used to remove the obvious nonground outliers. The track was then segmented into grids of length ΔL, where histogram peaks (W_g) were extracted in each grid following Gaussian smoothing. Grids were deemed valid only if the maximum peak $H_{wg(max)}$ fell within $[H_{min}, H_{max}]$ and the photon count exceeded N_1; otherwise, they were flagged as noise. For valid grids, DEM control points were extracted adaptively based on density. High photon count ($N > N_2$): To capture multipeak structures in complex terrains, the peak with the lowest elevation satisfying $W_g \geq \alpha W_{g(max)}$ was selected. Low photon count ($N \leq N_2$). The maximum peak $H_{wg(max)}$ was directly adopted as the control point.

2) DEM Control-Point Correction and Photon Classification: To mitigate anomalously high DEMs caused by missing ground points in the low-density circumstance ($N \leq N_2$), we performed a frequency-based statistical analysis. Outlier control points, defined by frequency $< \beta_1$ (frequency percentage threshold) and elevation $> H_{DEM(mean)}$ (global mean elevation), were removed and replaced by the mean of adjacent high-frequency points. Subsequently, a continuous DEM curve was generated via Gaussian smoothing and spline interpolation. Using a buffer of width Δd around this curve, photons were classified: those within the buffer as ground, those above the buffer but below the 95th percentile as vegetation, and the remainder as noise.

3) Refinement of Classification Results: Sole reliance on the buffer method creates robustness issues due to large-scale grids, causing misclassification of vegetation, particularly on steep mudflat surfaces, the edges of tidal creeks, and water points within the creeks. Therefore, targeted corrections were implemented as follows. First, in microgrids of width $\Delta L/5$, we compared the mean elevation difference between vegetation and ground points $[H_{v(mean)-g(mean)}]$. If this value was below ΔH_ρ, vegetation labels were considered unreliable and were corrected by relabeling vegetation points as ground. Where local ground points were unavailable, ground elevation was approximated using the mean ground elevation from adjacent windows. Tidal Creeks: a candidate set S was identified using frequency $<\beta_2$ and elevation $< H_{DEM(mean)}$. Subsequently, all DEM points lower than $\max(H_{DEM(i)})(i \in S)$ were retrieved. To prevent misidentifying bare mudflat slopes or low-lying depressions as creeks, regions exhibiting continuous monotonic trends or excessive continuous spans were excluded. Vegetation points within the validated creek regions and their adjacent windows (within $0.5\Delta L$) were cleared. Finally, to reduce isolated errors, vegetation or ground points were relabeled as noise if their counts within radius R fell below N_v or N_g, respectively.

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TABLE II
INFORMATION OF GENERAL PARAMETERS

$[H_{min}, H_{max}]$	ΔL	N_1	N_2	α	β_1
$[-2, 6]$	10 m	5	10	0.5	3%
Δd	ΔH_ρ	β_2	R	N_v	N_g
0.2 m	0.3 m	2%	50 m	10	15

The general parameters used in ICESat-2 photon denoising and classification are listed in Table II. However, due to the complex intertidal environment, parameters for specific ground tracks were adjusted based on local conditions. To ensure high-quality ground and vegetation points for subsequent model training, we conducted a detailed validation against optical imagery. This step guaranteed the accuracy and reliability of the classification.

B. Image Feature Construction

A total of 54 predictor variables were derived from the JL1-KF02 MS imagery, comprising 22 spectral and 32 texture features, to capture the spectral and spatial characteristics of intertidal land cover (Table III). The spectral features were designed to characterize the reflectance and absorption properties of surface materials across spectral regions and included: 1) single-band reflectance; 2) band ratios; and 3) established spectral indices. For each of the four bands (blue, green, red, and NIR), we extracted eight gray-level co-occurrence matrix (GLCM)-based texture metrics: mean (MEA), variance (VAR), correlation (COR), contrast (CON), dissimilarity (DIS), homogeneity (HOM), angular second moment (ASM), and entropy (ENT) [32]. By quantifying gray-level spatial distribution patterns within a pixel neighborhood, these metrics provide information closely associated with the structural characteristics of surface cover. Previous studies have demonstrated that utilizing these spectral and texture features has been shown to markedly improve the accuracy of mapping and elevation retrieval in intertidal zones [22], [33], [34].

Feature selection is essential to reduce redundancy and eliminate irrelevant predictors. In this study, we adopted a supervised feature selection strategy designed to mitigate multicollinearity while preserving the strongest predictive relationships with the target variable. Highly collinear feature pairs were identified using an absolute Pearson correlation threshold of |r| >= 0.98. This empirically selected, conservative threshold was used to remove near-duplicate predictors while avoiding excessive feature removal. For each collinear pair, we calculated the absolute correlation between each feature and the target variable; the feature with the stronger target relationship was retained, whereas the weaker one was discarded [35]. This process was iterated until no feature pairs exceeded the collinearity threshold. By combining correlation screening with target-guided selection, this approach systematically eliminates redundant information while prioritizing the most relevant predictors, thereby yielding an optimized and parsimonious feature set for model training.

C. Machine Learning Methods and Accuracy Evaluation

In this study, we employed four typical machine learning regression algorithms, RF, XGBoost, LightGBM, and SVM, to construct the inversion models [23], [24], [25], [26]. All models were implemented in the R environment. Following a stratified random sampling scheme, the photon-derived sample dataset was partitioned into training and testing subsets at a 7:3 ratio. Hyperparameters were optimized using Bayesian optimization with fivefold cross-validation on the training set, and model performance was evaluated on the test set. Model validation was further supported by field-measured vegetation heights for CHM accuracy and spatially continuous UAV-LiDAR datasets from four representative sites for DEM accuracy. Feature importance and driving mechanisms

TABLE III
FEATURES DERIVED FROM JL1-KF02 IMAGERY

Types	Features		Description
Spectral features	Single band reference: Blue, Green, Red, NIR		/
	Ratio index of two Band: NIR-Red (NR), NIR-Green (NG), NIR -Blue (NB), Red-Green (RG), Red-Blue (RB), Green-Blue (GB)		$\frac{\text{Band}_i}{\text{Band}_j}$ i, j = Blue, Green, Red, NIR, $i \neq j$
	Normalized difference vegetation index (NDVI)		$\frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$
	Renormalized difference vegetation index (RDVI)		$\frac{\text{NIR}-\text{Red}}{\sqrt{\text{NIR}+\text{Red}}}$
	Visible difference vegetation index (VDVI)		$\frac{2\times\text{Green}-\text{Red}-\text{Blue}}{2\times\text{Green}+\text{Red}+\text{Blue}}$
	Green normalized vegetation index (GNDVI)		$\frac{\text{NIR}-\text{Green}}{\text{NIR}+\text{Green}}$
	Chlorophyll index green (CIgreen)		$\frac{\text{NIR}}{\text{Green}}-1$
	Enhanced vegetation index2 (EVI2)		$2.5\times\frac{\text{NIR}-\text{Red}}{\text{NIR}+2.4\times\text{Red}+1}$
	Normalized green-red difference index (NGRDI)		$\frac{\text{Green}-\text{Red}}{\text{Green}+\text{Red}}$
	Normalized green-blue difference index (NGBDI)		$\frac{\text{Green}-\text{Blue}}{\text{Green}+\text{Blue}}$
	Red-green-blue vegetation index (RGBVI)		$\frac{\text{Green}^2-\text{Red}\times\text{Blue}}{\text{Green}^2+\text{Red}\times\text{Blue}}$
	Soil-adjusted vegetation index (SAVI) [1]		$\frac{(1+L)(\text{NIR}-\text{Red})}{\text{NIR}+\text{Red}+L}$
	Modified soil adjusted vegetation index (MASVI)		$\frac{2\times\text{NIR}+1-\sqrt{(2\times\text{NIR}+1)^2-8\times(\text{NIR}-\text{Red})}}{2}$
	Normalized difference water index (NDWI)		$\frac{\text{Green}-\text{NIR}}{\text{Green}+\text{NIR}}$
Textural features [2]	Blue	B_{MAE}, B_{VAR}, B_{COR}, B_{CON}, B_{DIS}, B_{HOM}, B_{SM}, B_{ENT}	$MAE = \sum_{i,j=0}^{L-1} P_{i,j} i$
	Green	G_{MAE}, G_{VAR}, G_{COR}, G_{CON}, G_{DIS}, G_{HOM}, G_{SM}, G_{ENT}	$VAR = \sum_{i,j=0}^{L-1} P_{i,j}(i-MEA)^2$
	Red	R_{MAE}, R_{VAR}, R_{COR}, R_{CON}, R_{DIS}, R_{HOM}, R_{SM}, R_{ENT}	$COR = \sum_{i,j=0}^{L-1} P_{i,j}\left[\frac{(i-MEA)(j-MEA)}{\sqrt{VAR\times VAR}}\right]$
	NIR	NIR_{MAE}, NIR_{VAR}, NIR_{COR}, NIR_{CON}, NIR_{DIS}, NIR_{HOM}, NIR_{SM}, NIR_{ENT}	$CON = \sum_{i,j=0}^{L-1} P_{i,j}(i-j)^2$
			$DIS = \sum_{i,j=0}^{L-1} P_{i,j}\lvert i-j\rvert$
			$HOM = \sum_{i,j=0}^{L-1} \frac{P_{i,j}}{1+(i-j)^2}$
			$SM = \sum_{i,j=0}^{L-1} \left(P_{i,j}\right)^2$
			$ENT = \sum_{i,j=0}^{L-1} P_{i,j}(-\ln P_{i,j})$

1 The L value in the SAVI ranges from 0 to 1. In this study, the commonly used value of 0.5 was adopted.
2 $P_{i,j}$ denotes the GLCM.

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were evaluated using Shapley additive explanations (SHAPs) analysis [36].

Model performance was assessed using the coefficient of determination (R^2), root-mean-squared error (RMSE), mean absolute error (MAE), and mean bias error (Bias), calculated as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (2)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (3)$$

$$\text{Bias} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i) \quad (4)$$

where y_i is the reference value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the reference values, and n is the number of samples used for evaluation. R^2 represents the proportion of variance in the reference data explained by the model, with values closer to 1 indicating better model performance. Both RMSE and MAE quantify the magnitude of prediction error, where smaller values indicate higher predictive accuracy. A positive Bias indicates overestimation, whereas a negative Bias indicates underestimation.

IV. Results

A. CHM Model Construction and Mapping

Following feature selection, 25 redundant or highly correlated predictors were removed, including 11 spectral features (RB, NB, NG, CIgreen, EVI2, GNDVI, NGBDI, NGRDI, RDVI, RGBVI, and SAVI) and 14 texture features (B_{MAE}, B_{VAR}, B_{CON}, B_{DIS}, G_{MAE}, G_{VAR}, G_{CON}, G_{DIS}, R_{MAE}, R_{VAR}, R_{CON}, R_{DIS}, NIR_{VAR}, and NIR_{CON}). The four regression models were trained with the remaining 29 predictors and achieved consistently good performance [see Fig. 4(a1)–(a4)]. RF and XGBoost achieved the best accuracy (R^2 = 0.62, RMSE = 0.40 m, and MAE = 0.29 m), followed by LightGBM (R^2 = 0.60, RMSE = 0.41 m, and MAE = 0.30 m), while SVM performed worst (R^2 = 0.57, RMSE = 0.43 m, and MAE = 0.31 m). Overall, the tree-based models better captured salt-marsh canopy structure.

The scatter-density plots [see Fig. 4(a1)–(a4)] show that most test samples were concentrated within the 1.0–2.0-m range, where the predicted values closely followed the 1:1 line. This indicates that the selected predictors could stably characterize canopy-height variations of the dominant vegetation. The residual trends [see Fig. 4(b1)–(b4)] showed negative slopes (−0.39 to −0.43), with intercepts greater than zero, indicating CHM overestimation for low-stature vegetation and a gradual shift toward underestimation as canopy height increased. The mean biases were close to zero [−0.02 to 0.01 m; Fig. 4(c1)–(c4)]. Specifically, RF showed a slight positive bias (0.01 m), XGBoost and LightGBM showed slight negative biases (−0.01 m), and SVM had the largest negative bias (−0.02 m). The residuals approximately followed a normal distribution with limited skewness, indicating that the CHM retrieval errors were generally stable in the complex salt-marsh environment.

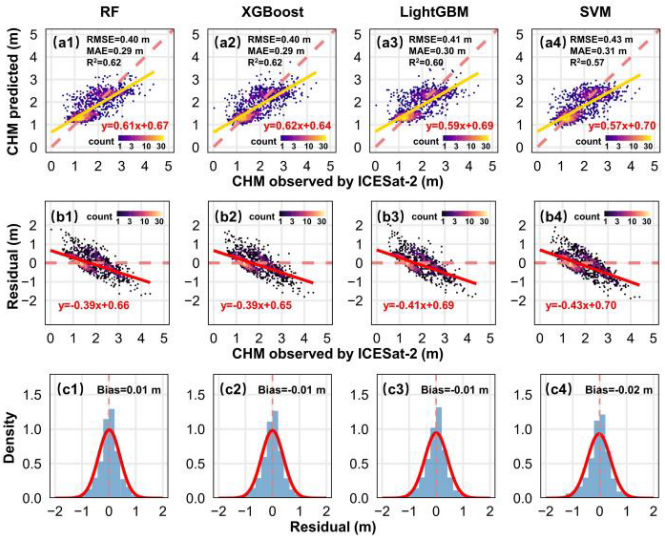


Fig. 4. Accuracy assessment of salt-marsh CHM prediction using RF, XGBoost, LightGBM, and SVM. (a1)–(a4) Scatter-density plots comparing model predictions with ICESat-2 observations. (b1)–(b4) Trends of residuals relative to observed CHM. (c1)–(c4) Statistical distributions of the residuals (red curves indicate the fitted normal distribution).

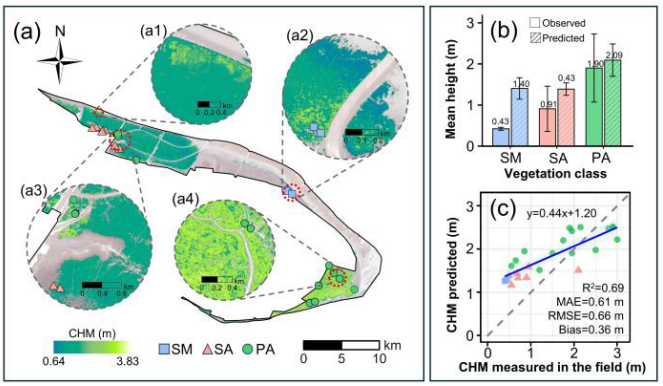


Fig. 5. Spatial mapping and field validation of salt-marsh CHM retrieved using the XGBoost model. (a) Spatial distribution of CHM across the study area, with insets (a1)–(a4) illustrating local spatial details in different vegetation zones. (b) Comparison of mean canopy heights between field samples and model predictions derived from the entire spatial retrieval at the community level. (c) Scatter plot validation of model predictions against independent field measurements (the solid line indicates linear regression, and the dashed line is the 1:1 line).

Using the optimal XGBoost model, canopy height was mapped across the entire study area. At the landscape scale [see Fig. 5(a)], the retrieved CHM showed clear spatial heterogeneity, with generally lower canopy heights in the north and higher canopy heights toward the southeast. This pattern was consistent with vegetation-community composition: the *Spartina alterniflora* (SA)-dominated northern area showed low CHM values (blue-green), whereas the *Phragmites australis* (PA)-dominated southeast area showed higher values (yellow-green). At local scales [see Figs. 5(a1)–(a4)], the CHM retained fine spatial texture. Figs. 5(a1) and (a3) illustrate mixed stands of PA and SA, where the CHM delineates boundaries between vegetation types and highlights the substantially taller canopy of PA. Fig. 5(a2), corresponding mainly to *Scirpus mariqueter* (SM), shows consistently low canopy

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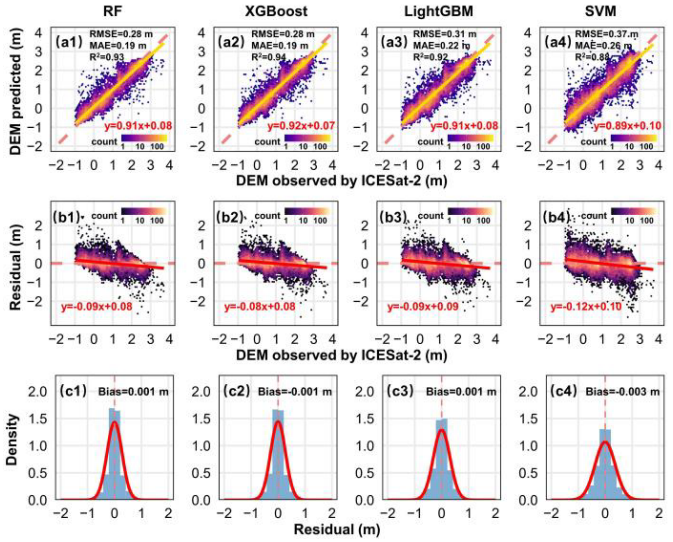


Fig. 6. Accuracy assessment of DEM prediction using RF, XGBoost, LightGBM, and SVM. (a1)–(a4) Scatter-density plots comparing model predictions with ICESat-2 observations. (b1)–(b4) Trends of residuals relative to observed DEM. (c1)–(c4) Statistical distributions of the residuals (red curves indicate the fitted normal distribution).

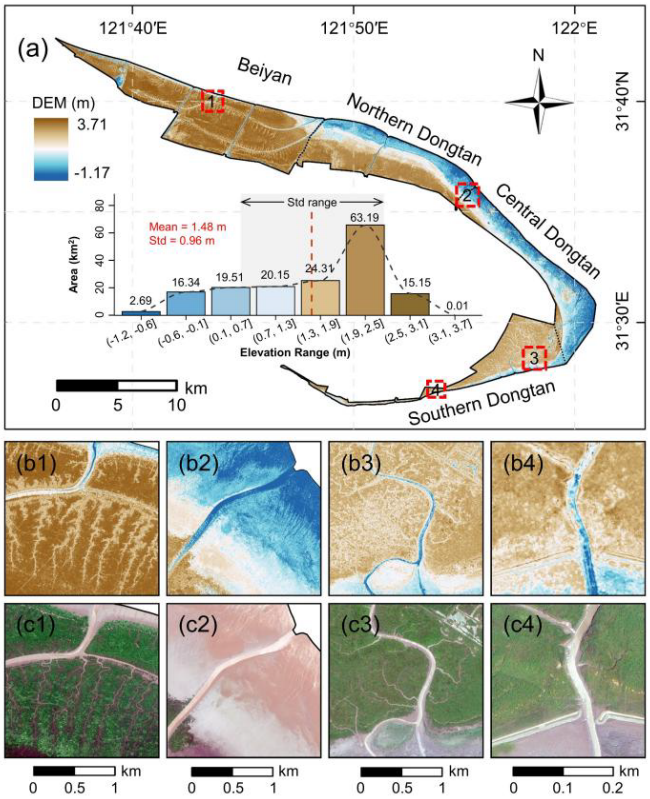


Fig. 7. Spatial distribution of the retrieved DEM and comparison with optical imagery. (a) 0.5-m DEM of the study area with elevation histogram. (b1)–(b4) Enlarged views of the retrieved DEM for four representative subregions. (c1)–(c4) Corresponding Jilin-1 true-color images.

heights with a clear seaward-decreasing gradient. Fig. 5(a4) focuses on a monospecific PA stand, where the CHM captured within-community microscale variability and visible textural detail.

Community-level comparisons [see Fig. 5(b)] revealed vegetation-dependent error patterns. Estimates for PA were the most accurate, with a retrieved mean height (2.09 m) closely matching the field mean (1.90 m). In contrast, the retrieved mean for SA (1.43 m) was substantially higher than the observed value (0.91 m). The largest discrepancy occurred for the shortest species, SM, which was markedly overestimated by approximately 0.97 m, indicating a pronounced positive bias.

Field validation [see Fig. 5(c)] showed overall agreement between predicted and observed heights (R^2 = 0.69, RMSE = 0.66 m, and MAE = 0.61 m). The large intercept of the regression model (1.20 m) indicates an elevated baseline, contributing to widespread overestimation in low-height areas. The regression slope was 0.44 and the intercept was 1.20 m [see Fig. 5(c)], indicating overestimation at low CHM values and underestimation at high CHM values. This pattern was generally consistent with the test set results (Fig. 4). Despite these systematic tendencies, the scatterplot displays a clear positive trend, indicating that the retrieved CHM preserves the relative rank order of vegetation heights. These results support the use of the CHM for characterizing the spatial distribution of vegetation-height classes across the complex intertidal landscape.

B. DEM Construction and Mapping

The predicted CHM was incorporated into the initial feature engineering as an auxiliary predictor when developing the DEM retrieval model. After feature selection, 26 redundant or highly collinear predictors were removed, including 14 spectral features (Blue, GB, RG, NB, NG, NR, CIgreen, EVI2, GNDVI, MSAVI, NDWI, RDVI, RGBVI, and SAVI) and 12 texture features (B_{MAE}, B_{VAR}, B_{CON}, G_{MAE}, G_{VAR}, G_{CON}, G_{DIS}, R_{VAR}, R_{CON}, R_{DIS}, NIR_{CON}, and NIR_{DIS}). Quantitative assessments [see Fig. 6(a1)–(a4)] indicated exceptional overall performance of the DEM retrieval models. XGBoost achieved the highest accuracy (R^2 = 0.94, RMSE = 0.28 m, and MAE = 0.19 m), followed closely by RF (R^2 = 0.93, RMSE = 0.28 m, and MAE = 0.19 m). LightGBM also maintained high accuracy (R^2 = 0.92, RMSE = 0.31 m, and MAE = 0.22 m), whereas SVM performed lowest but still demonstrated satisfactory performance (R^2 = 0.88, RMSE = 0.37 m, and MAE = 0.26 m). Scatter-density plots showed predictions tightly distributed around the 1:1 line, with fit slopes ranging from 0.89 to 0.92. Residual analysis [see Fig. 6(b1)–(b4)] further corroborated model stability. Residual-regression slopes were close to zero (−0.08 to −0.12), indicating minimal elevation-dependent bias and broadly uniform accuracy across the elevation gradient. Overall bias was negligible for all models [see Fig. 6(c1)–(c4)], and the error histograms exhibited sharp, symmetric Gaussian distributions centered at zero. This predominantly random-error structure indicates limited systematic bias and supports the robustness and reliability of the proposed stratified inversion strategy for DEM retrieval in complex intertidal environments.

Using the optimal XGBoost model, a 0.5-m resolution DEM was generated for the entire study area [see Fig. 7(a)]. Elevations ranged from −1.17 to 3.71 m, with a mean of 1.48 m and a standard deviation of 0.96 m. The elevation

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histogram was unimodal, peaking in the 1.9–2.5-m interval (~63.19 km^2). Most of the study area (~64.35%) fell within one standard deviation of the mean (0.52–2.44 m). Spatially, higher elevations formed continuous belt-like zones along the landward side adjacent to engineered seawalls and generally decreased from land to sea. However, geomorphic characteristics exhibited strong spatial heterogeneity across the region. The study area was subdivided into four zones: Beiyan, northern Dongtan, central Dongtan, and southern Dongtan. Subregional statistics showed that the vegetation-dominated zones (Beiyan and southern Dongtan) had higher mean elevations but lower variability (2.08 $\pm$ 0.64 m and 1.59 $\pm$ 0.59 m, respectively). In contrast, the bare-mudflat-dominated zones (northern and central Dongtan) had lower mean elevations but higher variability (1.27 $\pm$ 0.95 m and 0.43 $\pm$ 0.75 m, respectively). Further comparisons indicated that elevation patterns differed even within similar land-cover classes. Although Beiyan and southern Dongtan are both vegetated, Beiyan exhibited higher elevations with smoother relief, whereas southern Dongtan was lower and topographically more fragmented. Similarly, among the bare mudflat zones, northern Dongtan displayed a coherent land–sea elevation gradient, while central Dongtan—though generally flat—was characterized by extensive irregular microtopographic undulations and fragmented textures.

To assess the spatial consistency of the retrieved elevation, four representative subregions were selected and the retrieved DEM [see Fig. 7(b1)–(b4)] was compared with Jilin-1 true-color imagery [see Fig. 7(c1)–(c4)]. In densely vegetated salt marshes with well-developed tidal channels [Fig. 7(b1) and (b3)], the DEM reproduced channel-network morphology and clearly resolved elevation contrasts between low-lying creeks and relatively elevated vegetated platforms. The resulting texture and spatial patterns were consistent with the RGB imagery [see Fig. 7(c1) and (c3)]. Subaqueous topography in inundated creeks and nearshore shallow-water zones was also captured [see Fig. 7(b2)], where elevations were markedly lower than those of adjacent exposed mudflats. In addition, tidally driven microgeomorphic features were preserved [see Fig. 7(b2)], showing spatial structures consistent with hydro-geomorphic patterns indicated by the true-color imagery [Fig. 7(c2)].

Quantitative validation using UAV-LiDAR data was conducted for four independent regions (Fig. 8). R^2 values were 0.77, 0.63, 0.64, and 0.53, and RMSE ranged from 0.34 to 0.53 m. High-density point clusters were tightly distributed around the 1:1 reference line, indicating good agreement between retrieved elevations and LiDAR observations for most pixels. Bias remained low (−0.10 to 0.16 m): Region (I) exhibited a positive bias, Region (IV) showed a slight negative bias, and Regions (II) and (III) were nearly unbiased. Scatter patterns exhibited clear elevation dependence. At low elevations (<1 m), points in Region (I) closely followed the reference line, whereas Regions (II)–(IV) showed greater vertical dispersion. At intermediate elevations (1–2 m), relative dispersion among regions differed from that in the low-elevation range. At elevations > 2 m, Regions (I) and (II) showed the lowest dispersion and most closely followed the 1:1 trend, Region (III) showed larger dispersion, and Region (IV) exhibited the greatest spread with a tendency toward

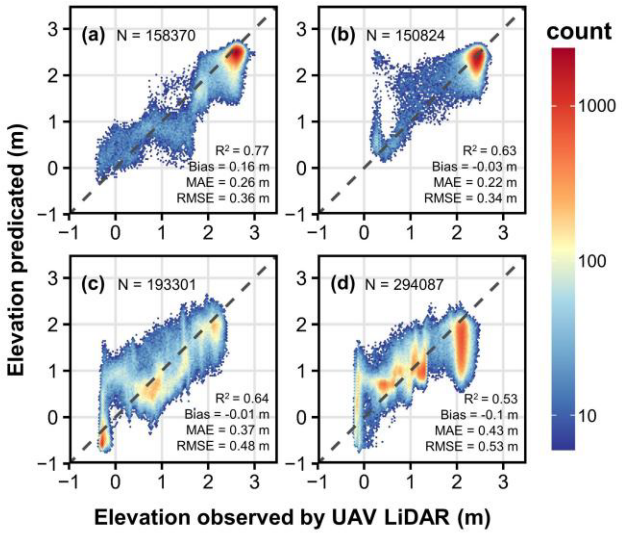


Fig. 8. Validation of the retrieved DEM against UAV-LiDAR elevations for four independent sites. (a)–(d) UAV-LiDAR regions (I)–(IV), respectively.

underestimation. Overall, the retrieved DEM achieved high accuracy across all four validation regions, although error magnitude and its elevation dependence varied among regions.

V. Discussion

A. Feature Importance and Contribution Analysis

SHAP revealed distinct feature-attribution structures across algorithms and surface environments (Fig. 9). Tree-based models (RF, XGBoost, and LightGBM) showed more concentrated feature contributions, with the top 8–12 predictors reaching 80% cumulative contribution in the full domain and over bare mudflats. In contrast, SVM required 17–18 predictors to reach the same threshold, indicating a more distributed multivariate representation. In vegetated areas, the number of predictors required to reach 80% cumulative contribution became more similar across models (approximately rank 14), suggesting that the models relied on broadly similar information sources under canopy-covered conditions.

Subregional SHAP patterns showed that the dominant predictors differed between bare mudflats and vegetated areas. Over bare mudflats, elevation retrieval relied mainly on spectral predictors, including the Green band, NDVI, and RB ratio, likely because surface moisture and sediment conditions vary along the elevation gradient and influence optical reflectance. The sign changes in the SHAP values of these predictors further indicate context-dependent effects, reflecting the nonlinear relationship between spectral response and elevation in intertidal environments. In vegetated areas, feature importance shifted toward a complementary spectral-texture-CHM structure. Texture metrics increased in rank, suggesting that spatial texture captured part of the canopy-structure and patch-heterogeneity information. CHM became a stable mid-to-high ranked predictor, typically ranking fourth–seventh, indicating that canopy-height information provided useful complementary information for DEM retrieval in vegetated areas. More specifically, high CHM values were generally associated with negative SHAP contributions in the DEM model. In vegetated

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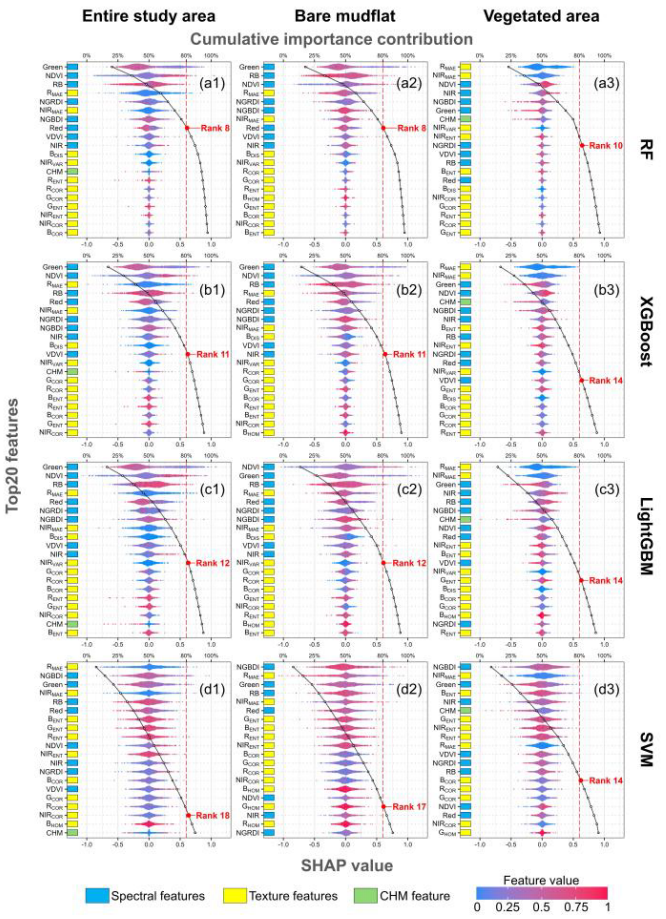


Fig. 9. Feature contribution structures for DEM inversion across three scenarios. Columns (1)–(3) indicate the three surface settings, and rows (a)–(d) indicate the four models. Panel IDs are given as (a1)–(d3). Each panel shows SHAP value distributions and the cumulative contribution curve (black); the red marker indicates the feature rank at the 80% cumulative contribution threshold.

salt marshes, spectral and texture features may mix signals from the canopy and the underlying ground surface, whereas CHM provides an explicit canopy-height constraint. This negative contribution suggests that the model used CHM to reduce predicted ground elevation where canopy-height effects were stronger, thereby compensating for potential canopy-related elevation offsets and supporting canopy-terrain decoupling.

B. Performance and Error Drivers in Complex Intertidal Environment

Retrieved elevations validated against UAV-LiDAR (Fig. 10) demonstrate that the proposed synergistic approach accurately reconstructs intertidal topography, capturing regional gradients and intricate tidal-channel networks. Accuracy remained robust (R^2 = 0.53–0.77 and RMSE = 0.34–0.53 m), with median errors fluctuating around zero. Residual maps [see Fig. 10(c1)–(c4)] and elevation-binned boxplots [see Fig. 10(d1)–(d4)] revealed elevation-dependent error and spatial heterogeneity, with overestimation at low elevations and underestimation at higher elevations.

Overestimation mainly occurred near tidal-channel banks and may be associated with complex surface composition and sensing limitations along channel margins. Tidal-channel

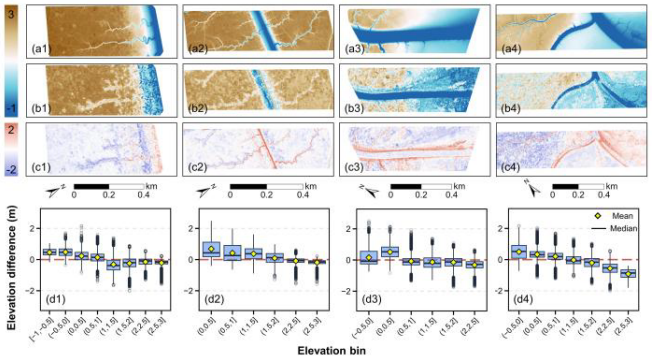


Fig. 10. Validation of the retrieved DEM against spatially continuous UAV-LiDAR DEM across four sites. (a1)–(a4) UAV-LiDAR reference elevation. (b1)–(b4) Retrieved DEM. (c1)–(c4) Elevation difference (retrieved DEM minus UAV-LiDAR reference elevation). (d1)–(d4) Boxplots of elevation differences binned by reference elevation for each site.

edges are complex zones where flats, water, and vegetation may coexist, and mixed spectral signals can still occur even in submeter imagery. In addition, specular reflection from wet channel banks under certain viewing geometries may produce spectral anomalies, while ICESat-2 sampling in these areas may yield fewer valid returns or reduced representativeness. Sparse training samples, together with strong microtopographic heterogeneity, may make it difficult for the model to capture local high-frequency terrain variations, thereby causing localized bias.

In contrast, underestimation mainly occurred in higher elevation areas, particularly within bare-flat-to-vegetation transition zones or inside vegetated areas, and showed regional differences. Retrievals were relatively stable in the northern sites (Sites I and II), whereas underestimation was stronger in the southern sites (Sites III and IV), particularly at Site IV. Consistent with the feature contribution patterns in Section V-A, this bias may be related to the negative contribution of CHM and uncertainty in CHM prediction. High CHM values in tall vegetated areas may lead the model to apply stronger downward adjustment to predicted ground elevation; if CHM is locally overestimated, especially over low-stature vegetation or transition zones, this adjustment may further amplify DEM underestimation. Species and community structure may also contribute to this uncertainty because different salt-marsh communities can affect the vertical detectability and separability of ATL03 vegetation and ground photons and may alter the relationship between CHM and ground elevation. Therefore, although CHM provides a useful structural constraint for canopy-terrain decoupling, CHM uncertainty and community-related structural differences may still propagate into DEM retrieval. However, this propagation should be interpreted as a potential error pathway rather than a deterministic one-to-one transfer, because DEM retrieval is jointly controlled by CHM, spectral predictors, texture metrics, and their nonlinear interactions. Notably, although local vegetated areas showed systematic underestimation, their interquartile ranges were generally smaller than those of bare flats, indicating lower error dispersion and smaller overall variability in these areas.

C. Methodological Innovation and Comparative Analysis

This study developed an intertidal wetland topographic retrieval framework based on ATL03-derived DEM/CHM

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TABLE IV
COMPARISON OF REPRESENTATIVE OPTICAL-IMAGE-BASED INTERTIDAL DEM RETRIEVAL METHODS

Method	Data requirement	Spatial resolution	Applicability in vegetated areas	Main limitation
Waterline method	Multi-temporal optical images acquired at different tidal stages; tide-level data	Depends on image resolution, waterline density, and interpolation	Limited; prone to canopy obstruction and reduced reliability in vegetated areas.	Not applicable to vegetated areas; relies on multi-temporal imagery; interpolation resolution is relatively low.
Inundation-frequency method	Dense long-term optical image records; elevation calibration/control data	Typically 10–30 m	Limited; prone to canopy obstruction and reduced reliability in vegetated areas.	Not applicable to vegetated areas; relies on large volumes of historical observation imagery; sensitive to the distribution of observations across tidal levels.
Existing machine-learning retrieval methods	Single-phase/composite medium-resolution optical imagery; ICESat-2 DEM samples	Typically 10–30 m	Partly applicable; canopy-terrain decoupling remains limited	Requires single-phase high-quality imagery; lacks explicit vegetation-structure constraints.
This study	Single-phase high-resolution optical imagery; ICESat-2 CHM/DEM samples	0.5 m	Applicable; designed for vegetated areas through CHM constraint	Requires single-phase high-quality imagery and high-quality ICESat-2 sample extraction.

sample construction, single-phase 0.5-m high-resolution optical imagery, and CHM-based structural constraints. Its key methodological advantage is the ability to retrieve high-resolution intertidal topography, including vegetated salt-marsh areas, without relying on dense long-term optical image time series.

Existing optical-image-based intertidal DEM retrieval methods are commonly constrained by their dependence on multitemporal observations and their limited applicability in vegetated areas. Waterline methods require waterlines extracted from images acquired at different tidal stages and depend on accurate tide-level matching and interpolation between waterlines [17]. Inundation-frequency methods rely on dense long-term image records and must account for the uneven distribution of observations across tidal levels when estimating inundation frequency [18]. Both methods essentially depend on dynamic observations of water extent and can be effective over bare mudflats. However, in vegetated areas, their reliability decreases or may even fail because waterlines or inundation boundaries can be obscured by canopy cover. In addition, dense long-term high-resolution optical image sequences are often difficult to obtain, further limiting their application in high-resolution intertidal topographic mapping.

In our previous work, we attempted to combine ICESat-2 photon elevation samples with medium-resolution optical image features and used machine learning models to establish relationships between elevation samples and image-derived predictors [21], [22]. However, in vegetated areas, optical features mainly represent canopy-surface information and lack explicit structural constraints linked to true ground elevation. Meanwhile, the spatial resolution of medium-resolution imagery limits its ability to characterize fine tidal-creek structures and microtopographic variations. By contrast, the proposed framework relies only on single-phase 0.5-m optical imagery and ICESat-2 ATL03 samples, enabling DEM retrieval at 0.5-m spatial resolution and extending its applicability from primarily bare mudflats to vegetated salt-marsh areas. By introducing CHM as a structural constraint, the framework explicitly represents canopy-height information and enhances the model's ability to decouple canopy signals from true ground elevation. To further clarify the methodological differences, Table IV compares representative optical-image-driven intertidal DEM retrieval methods in terms of data requirements, spatial resolution, applicability in vegetated areas, and main limitations.

The basis for achieving these advantages is accurate and efficient ICESat-2 photon classification. Results from representative areas (Fig. 11) show that the proposed photon-classification method performs consistently across bare mudflats, vegetated areas, and tidal creeks, effectively separating ground and vegetation photons. This provides reliable inputs for CHM and DEM retrieval. To maintain sample purity, this study adopted relatively stringent photon filtering and denoising procedures. Although this process may remove some uncertain photons in sparsely vegetated areas or near tidal-creek margins, it helps improve sample purity and provides higher-quality training data for subsequent machine learning-based retrieval.

D. Limitations and Future Directions

Although the proposed framework can generate continuous topographic information across bare flats, salt marshes, and shallow-water areas using single-phase high-resolution imagery, it remains constrained by the characteristics of the input data. First, in a single-phase setting, spectral and textural signals may be affected by transient surface conditions, such as tide stage, waves, specular reflection, and shadows. In addition, while high-resolution imagery provides finer spatial texture information, it may also introduce greater local heterogeneity and nontopographic noise, making feature-elevation mapping more complex. High-resolution retrieval thus necessitates stricter quality control and robust feature selection strategies. Meanwhile, the along-track linear sampling pattern of ICESat-2 limits its representation of complex geomorphic units, and the along-track photon spacing of approximately 0.7 m places higher requirements on sample continuity when combined with submeter imagery. In steep tidal-creek banks and vegetation edges, water interference, slope-induced scattering, and canopy obstruction may reduce the number of effective samples. In sparse or low-stature vegetation areas, limited sample availability and weaker canopy-ground separability may further introduce retrieval uncertainty.

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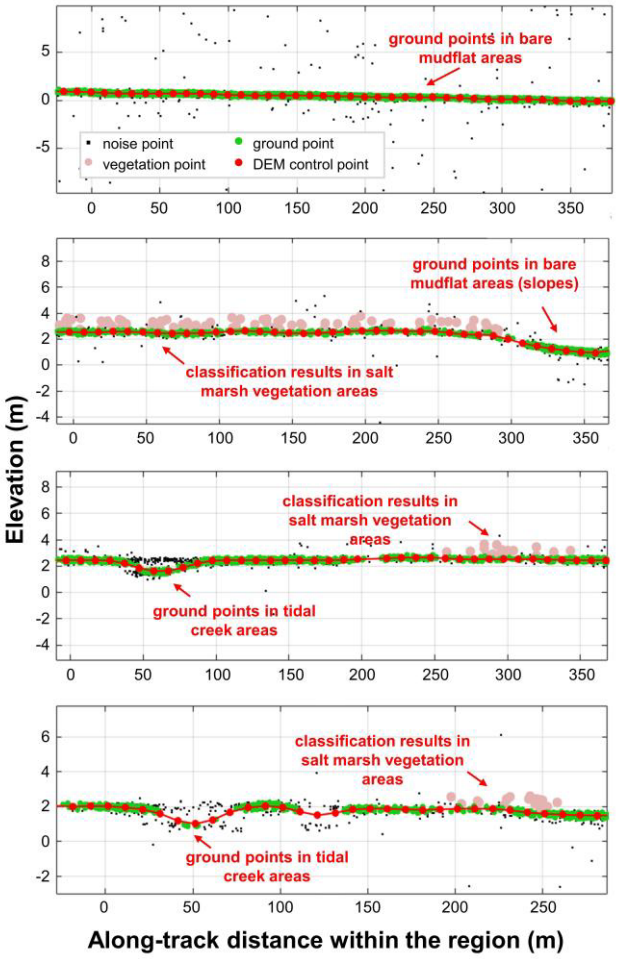


Fig. 11. ICESat-2 photon classification in typical intertidal environments using the proposed method.

In addition, acquisition-time differences among the multisource datasets may introduce uncertainty. For model construction, the Jilin-1 image was acquired in July 2024, while the DEM and CHM samples were derived from ATL03 photons within approximately one-year and three-month temporal windows, respectively. These windows were defined to balance sample availability, spatial coverage, and temporal consistency, but uncertainties arising from acquisition-time differences cannot be fully avoided. For accuracy validation, CHM performance was assessed using field vegetation-height measurements collected during the same growing season, whereas DEM accuracy was independently evaluated using UAV-LiDAR datasets acquired in December 2024 and April 2025 across four representative regions. Although the UAV-LiDAR datasets were selected to be as close as possible to the satellite observation period, they were not strictly synchronous. The UAV-LiDAR time lag may introduce validation uncertainty because tidal forcing, hydrodynamic erosion and deposition, typhoons, storm-surge events, and local human disturbances can alter intertidal topography over short time scales. In vegetated salt marshes, strict filtering was applied to remove vegetation returns and extract ground points; nevertheless, phenological differences in vegetation height, density, lodging, and canopy structure may still affect data acquisition and ground-point extraction, thereby influencing the validation results.

Spatial autocorrelation among remote sensing samples may affect accuracy assessment on the internal test set. Nearby pixels and photon-derived samples often share similar spectral, textural, and geomorphic conditions. Therefore, random or target-stratified splitting may place spatially neighboring and similar samples into both the training and testing subsets, potentially leading to an overestimation of model accuracy. Although field-measured vegetation heights and UAV-LiDAR datasets provide external reference data for CHM and DEM validation, respectively, future work could consider spatial-block partitioning strategies when denser and more spatially distributed training and validation datasets become available.

Another limitation is that species- and community-level vegetation effects were not explicitly represented in the model. Previous studies have shown that vegetation species composition, structural traits, and interpretable modeling are valuable for improving vegetation remote sensing retrieval [37], [38], [39]. In this study, species labels were not included during model construction because a reliable wall-to-wall salt-marsh species-classification map was unavailable. Instead, CHM, texture, and spectral features were used to indirectly represent species- and community-related structural differences. Different vegetation communities may differ in ATL03 canopy-ground separability and spectral-saturation effects. This design reduced dependence on additional species-mapping products, but it also limited our ability to explicitly distinguish species-specific error sources. Future work could integrate reliable salt-marsh species or community classification results to further examine species-specific CHM errors and their propagation into DEM retrieval, thereby improving the interpretability of canopy-constrained DEM retrieval. The role of CHM constraints may also vary with spatial scale, because image resolution can affect the level of detail in surface-feature representation, image noise, effective sample size, and the relative contributions of different predictors. Future work should further evaluate the stability of the framework across different spatial resolutions and its influence on DEM accuracy.

VI. Conclusion

This study developed a canopy-height-constrained framework for high-resolution DEM retrieval in complex intertidal wetlands by integrating ATL03-derived DEM/CHM samples, single-phase 0.5-m optical imagery, and machine learning. By deriving DEM and CHM samples directly from ATL03 photon classification and introducing CHM as a structural constraint, the framework links photon-level canopy-ground separation with wall-to-wall optical prediction. The retrieved 0.5-m DEM reproduced regional elevation gradients, tidal-creek networks, and microtopographic variations across bare flats and vegetated salt marshes, with accuracy supported by independent UAV-LiDAR validation. SHAP analysis further showed that CHM provided useful structural information for canopy-terrain signal decoupling in vegetated areas. These results indicate that the framework offers a feasible pathway for high-resolution intertidal DEM retrieval where dense long-term optical image sequences are difficult to obtain. Nevertheless, CHM uncertainty, sparse ATL03 sampling,

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acquisition-time differences among multisource datasets, and vegetation-community effects may influence DEM retrieval and validation. Future work should test the framework across broader geomorphic and vegetation settings and further quantify the propagation of CHM uncertainty into DEM retrieval.

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